

ESS Investigation

Earth's Systems

The Ripple Effect — The Work of Water Across New York State Surfaces

Teacher Materials

Investigation Overview

In this investigation, students will collaboratively plan and conduct an investigation of the properties of water and its effects on Earth materials and surface processes occurring in New York State. Pairs of students will plan and conduct two investigations on three surface processes to gather measurable (quantitative) data to be used as evidence for the effects of water on Earth surface processes. Student groups will be provided with specific criteria and constraints in order to plan and conduct their chosen Earth surface process investigation. Each group will communicate their data to the rest of the class for individual analysis.

In **Session 1: Collaborative Student Work**, students will collaborate in pairs to plan and conduct one of two **Investigation Options (A or B)** on the properties of water and its effects on Earth materials and surface processes. All students will also complete **Investigation C**.

In **Part 1**, students will collaborate in pairs to:

- plan an **Investigation Option, A or B**
- identify the phenomenon (variable) to be investigated for their chosen option (a connection between the properties of water and its effects on Earth materials and surface processes)
- make a prediction about how the identified phenomenon is affected by water
- develop a written investigation plan that describes the data that will be collected and how the data is relevant for determining the effect of water on Earth materials and surface processes, based on given criteria and constraints

In **Part 2**, students will:

- continue to collaborate in pairs to conduct their chosen investigation
- gather measurable (quantitative) data to be used as evidence for the effects of water on Earth materials
- communicate their data to the rest of the class for analysis
- perform **Investigation C**
- record data from the other **Investigation Option**

In **Session 2: Individual Student Work**, students will independently answer the investigation questions in ***Student Answer Packet 2***.

Standards Alignment

Primary PE

HS-ESS2-5: Plan and conduct an investigation of the properties of water and its effects on Earth materials and surface processes.

Related PEs

HS-ESS2-1: Develop a model to illustrate how Earth’s internal and surface processes operate at different spatial and temporal scales to form continental and ocean floor features.

HS-ESS2-2: Analyze geoscience data to make the claim that one change to Earth’s surface can create feedbacks that cause changes to Earth’s systems.

Primary DCI

HS-ESS2.C: The Roles of Water in Earth’s Surface Processes: The abundance of liquid water on Earth’s surface and its unique combination of physical and chemical properties are central to the planet’s dynamics. These properties include water’s exceptional capacity to absorb, store, and release large amounts of energy, transmit sunlight, expand upon freezing, dissolve and transport materials, and lower the viscosities and melting points of rocks.

Related DCIs

ESS2.A: Earth Materials and Systems: Earth’s systems, being dynamic and interacting, cause feedback effects that can increase or decrease the original changes. (HS-ESS2-1) (HS-ESS2-2)

ESS2.B: Plate Tectonics and Large-Scale System Interactions: Plate movements are responsible for most continental and ocean-floor features and for the distribution of most rocks and minerals within Earth’s crust. (HS-ESS2-1)

Primary SEP

Planning and Carrying Out Investigations: Plan and conduct an investigation individually and collaboratively to produce data to serve as the basis for evidence, and in the design: decide on types, how much, and accuracy of data needed to produce reliable measurements and consider limitations on the precision of the data (e.g., number of trials, cost, risk, time), and refine the design accordingly. In this investigation, students will plan and carry out a chosen investigation related to water cycle processes and the effect of water on these processes. The effect of water on Earth systems emphasizes mechanical and chemical investigations related to permeability, runoff, deposition, infiltration, and water retention. In addition, all students will carry out an investigation related to the weathering of an Earth material by a water cycle process.

Related SEPs

Developing and Using Models: Develop a model, based on evidence, to illustrate the relationships between systems or between components of a system. (HS-ESS2-1) In this investigation, students will develop and use models to collect data to examine large-scale Earth surface processes on a smaller spatial and temporal scale to determine the effect of water on Earth systems.

Analyzing and Interpreting Data: Analyze data, using tools, technologies, and/or models (e.g., computational, mathematical), in order to make valid and reliable scientific claims or to determine an optimal design solution. (HS-ESS2-2) In this investigation, students will analyze data to determine the amount of erosion, deposition, runoff, permeability, and water retention related to rock material within different Earth systems.

Primary CCC

Structure and Function: The functions and properties of natural and designed objects and systems can be inferred from their overall structure, the way their components are shaped and used, and the molecular substructures of its various materials. In this investigation, systems such as stream tables and rain gardens are used to determine the effects of water on Earth systems and processes. Additionally, the effect of different liquids on the weathering of samples of chalk will be studied.

Related CCCs

Stability and Change: Change and rates of change can be quantified and modeled over very short or very long periods of time. Some system changes are irreversible. (HS-ESS2-1) In this investigation, students will quantify amounts and rates of change using stream tables, rain gardens, and various rock and mineral sediments in order to determine the effect of water on Earth systems.

Stability and Change: Feedback (negative or positive) can stabilize or destabilize a system. (HS-ESS2-2) Students will identify feedback systems within water cycle processes.

Teacher Directions

Preparing the Investigation:

This investigation, broken into **Session 1: Part 1** (Planning the Investigation), **Session 1: Part 2** (Conducting the Investigation), and **Session 2**, is intended to satisfy a maximum of 160 minutes toward the 1200 minutes of hands-on laboratory experience required by Part 100 of the Commissioner’s Regulations. Investigations must be documented as satisfactory to be counted toward the 1200-minute requirement.

Appropriate safety practices are left to local discretion. The teacher should communicate the appropriate use and care of safety equipment and the associated procedures to students.

Determine how many station setups you will need, knowing that students will be working in pairs for **Part 1** and **Part 2** of the investigation. For example, if you have 30 students, you will need a minimum of eight setups for **Investigation Option A**, a minimum of eight setups for **Investigation Option B**, and a minimum of 15 setups for **Investigation C**.

In **Session 1 (Part 1)**, assign students to pairs. In ***Student Answer Packet 1***, students will collaborate in pairs to create a written investigation plan. Using models to collect data, students will examine one large-scale Earth surface process on a smaller temporal and spatial scale in order to provide evidence for the effect of water on Earth materials and surface processes. Students will identify the dependent and independent variables and make a prediction that specifies what happens to the dependent variable when the independent variable is manipulated.

Students have the opportunity to complete one of two **Investigation Options (A or B)** to plan and conduct. However, data for **BOTH Investigation Options** need to be recorded in ***Student Answer Packet 1***.

Before the procedure begins, instruct students to read through **Investigation Options A, B, and C**, found on pages 3 through 12 of the ***Student Directions***, in order to make a choice between **Investigation Options A or B**.

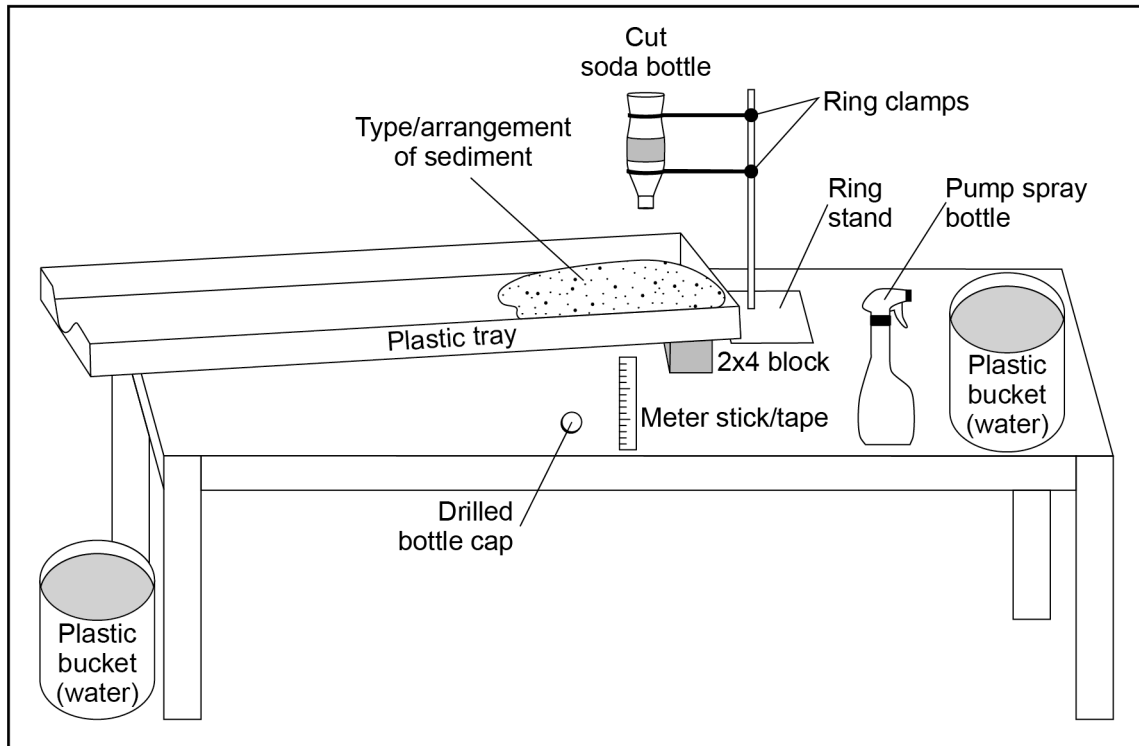
The teacher should make sure that in each class BOTH Investigation Options have been conducted by at least one pair of students.

Additionally, all students will collaboratively complete **Investigation C**.

In **Session 2**, students will individually complete ***Student Answer Packet 2***.

Part 1 Investigation Options:**Option A: Stream Table**

Students will plan and conduct an investigation, using materials listed under the model, to produce data to be used as evidence for the effect of water on Earth's materials and surface processes, specifically erosion and deposition.

Model of Stream Table Setup**List of Possible Materials**

• stream table or plastic (under the bed) storage tote 41 qt. with a 5/8-in. hole drilled in the center of one short side 1/2 in. from the bottom • washcloth (to simulate a vegetated surface) • acetate sheet (to simulate an urban surface) • meter stick/tape	• 2-L soda bottle with the bottom cut off and a 1/4-in. hole drilled in the cap or 16-oz plastic cup with evenly distributed holes in bottom • fine sand – wet • coarse sand – wet • gravel (pebble-sized particles) – wet • figurines* (like Monopoly houses) • ring stand and ring clamp	• 2x4 blocks of wood • collection bucket (2-2.5 gal. bucket/paint bucket) • water • spray bottle (for wetting sediment) • stopwatch • acetate sheets – cut lengthwise (to be used to divide stream table)
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* For the purpose of producing quantitative data

Criteria/constraints for students completing Investigation Option A:

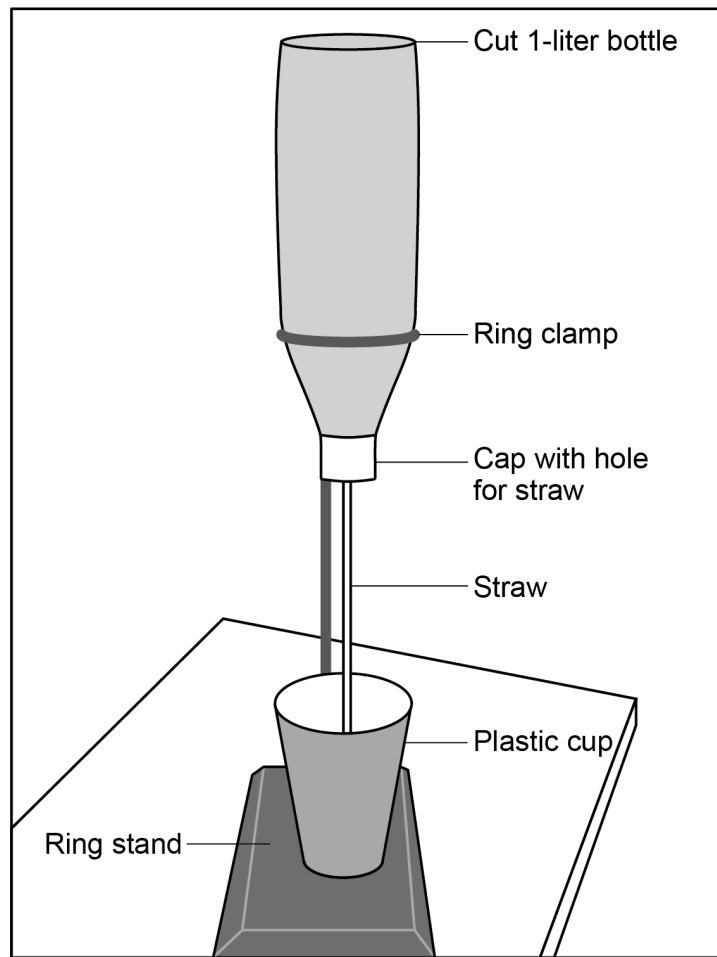
- can use one size of sediment alone *or* a mix of different-sized sediments (e.g. 75% fine sand and 25% gravel or 50% medium sand and 50% gravel)
- inform students that they should only use enough sediment to create a landscape that allows for quantitative evidence of erosion and deposition (e.g. by measuring the distance particles/figurines move)
- it is recommended that four waste sediment containers (fine sand, coarse sand, gravel, mixed) be available and clearly labeled for students to place used sediment in
- it may be necessary to support the underside of the stream table model with rigid material such as plywood

Option B: Rain Garden

Students will plan and conduct an investigation, using materials listed under the model, to test the effect of different materials on permeability, infiltration, runoff, and water retention in a rain garden design. The data will be used as evidence to determine which materials should be considered in effective urban rain garden designs.

The chosen materials to be placed in the 1-liter bottle are not shown in the model.

Model of Rain Garden Setup



List of Possible Materials

• 1-L clear soda bottle with the cap drilled to the diameter of a straw • 16-oz. bottle of any type – cap predrilled with 1/8-in. holes (to simulate a rain shower) • straw • ruler	• coffee filter • cheese cloth • cotton balls • stopwatch • ring stand and clamp • food coloring (blue or red) • 100-ml graduated cylinder • spoons	• paper towels • beakers • gravel (pebble-sized particles) – wet • fine sand – wet • coarse sand – wet • potting soil – wet
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Criteria/constraints for students completing Investigation Option B:

- should not exceed 400 to 500 ml of sediment in the bottle (one full 16-oz. plastic cup)
- all sediments should be wet before conducting the investigation
- colored water can either be prepared ahead of time by the teacher **or** by students
- only enough food coloring drops to achieve an observable color change in the water should be used (as this will ensure enough food coloring for other classes)
- the top of the straw must not extend into the bottle
- no sediment should be transported into the bottom cup as the focus of the investigation is flow of water, not sediment
- the rain simulator may need to be tilted or vented to achieve a continuous flow of water into the rain garden

Investigation C: Weathering of Earth Materials

Students will conduct an investigation, using materials listed under **Investigation C**, to collect data to be used as evidence on the chemical effects of tap water (precipitation) and vinegar (acid rain) on chalk (rocks) in New York State. Students will determine how these processes contribute to the formation of the region's unique geological features.

In **Session 1: Part 2**, students will gather the materials necessary to conduct their planned investigation. In **Student Answer Packet 1**, students will record their own data and make a claim about how the collected data supports or refutes their prediction made in **Session 1: Part 1**. All students will use the materials for **Investigation C**.

List of Possible Materials

- chalk – two pieces (approximately 2.5 - 3.5 cm in length)
- white vinegar
- 3.5 oz. transparent plastic cups – two
- electronic balance
- masking tape
- tap water
- stopwatch
- goggles
- tweezers

All three investigations will require general materials for cleanup (paper towels, mop, broom, etc.)

Preparation for Session 1: Part 1 (Planning the Investigation)

- 1) Assemble materials for the number of students in your class. (***Student Directions*** and ***Student Answer Packet 1*** should be copied and ready for distribution.)
- 2) All materials for both **Investigation Options** should be available for students to assist in planning their investigation. It is recommended that the materials be separated by **Option A** and **B** and labeled accordingly.

Preparation for Session 1: Part 2 (Conducting the Investigation)

- 1) Have a plan for the method by which **Investigation Option A** or **B** student data is shared.
- 2) Make a large, reusable class data table for **Investigation C** so that students can record and post their data in a location determined by the teacher.
- 3) Replenish any materials used from prior classes to their initial **Investigation Option** amounts before the next class.

Preparation for Session 2

- 1) Make sure ***Student Answer Packet 2*** has been copied and is ready for distribution to each student.
- 2) Return ***Student Answer Packet 1*** and ***Student Directions*** to students as they could be used for reference in completing ***Student Answer Packet 2***.

Student Directions with Teacher Annotations

Focus Questions: While performing this investigation, use the following focus questions to guide your work.

1. How is Earth's surface changed through erosional and depositional processes?
2. How can measurements of permeability of different materials be used as evidence for the predicted amount of infiltration, surface runoff, and retention of water?
3. How does water, in its different forms, chemically and mechanically cause changes in New York State rocks?
4. How can water cycle processes change Earth's surface and create feedbacks that cause changes to Earth systems?
5. How can models be used to illustrate water's effects on surface processes that operate at different spatial and temporal scales?
6. What effects do the properties of water have on Earth materials and surface processes?
7. How does water affect the melting points of rocks?

Inform students that they are not answering the focus questions at this time. The focus questions are included as an overview of concepts related to this investigation. These questions will be addressed through the students' performance of this investigation.

Procedure:

Session 1: Part 1 (Planning the Investigation)

1. Write your name on ***Student Answer Packet 1***.
2. Read and review the information about the **Investigation Options (A and B)**.

It is strongly recommended that there is an equal number of groups investigating each option. It might be necessary to assign groups to the **Investigation Options**.

3. In ***Student Answer Packet 1***, indicate the letter of the **Investigation Option** that you and your partner are conducting.
- 4a. In ***Student Answer Packet 1***, identify the independent and dependent variables.
- 4b. Make a prediction that specifies what happens to the dependent variable when the independent variable is manipulated.

It is strongly recommended that a list of potential independent variables to investigate (found in ***Student Answer Packet 1 Answer/Scoring Key***) be generated in a whole-class discussion. This list will reduce the amount of time spent planning by the students. It is expected that, after choosing an independent variable, students should be able to identify the appropriate data to be collected (dependent variable). Remind students that standard experimental design includes controlling variables to establish validity and to determine cause-and-effect relationships.

As students are identifying variables, circulate around the room to ensure students are not misidentifying variables in the planning phase of the investigation.

5. With your partner, plan an investigation to determine the effect of your independent variable on Earth's materials and surface processes. Be sure to control all variables except the one you are investigating and perform three trials for each value of the independent variable.

Criteria: Your Investigation Option Plan must include...

- a title
- the purpose of the investigation
- a description of the numerical data, including units, to be collected
- evidence derived from the data
- a description of how the data collected will be relevant to determining the effect of water on Earth materials and surface processes
- a selection of appropriate tools/materials to collect, record, analyze, and evaluate data
- a step-by-step procedure that must be focused on the purpose of the investigation and be specific enough so that other pairs of students can conduct the exact same investigation by reading your procedure
- a labeled model of your chosen investigation setup
- a student-designed data table to record numerical data

Constraints:

- you are limited to the materials provided for each option
- your investigation should be completed in approximately 40 minutes (one class period)
- you **may not** taste any substance

6. Create an investigation plan (a written, step-by-step procedure), based on the criteria and constraints.

Each student needs to record the investigation plan in their own **Student Answer Packet 1**.

7. Create your data table in **Student Answer Packet 1**. Include a title, dependent variable, independent variable, three trials, averaged data, and any observations.

If students need more space than what is provided, instruct students to attach their data table to the back of **Student Answer Packet 1**.

If **Session 1: Part 2** is occurring on another day, collect all the student materials so that they can be returned for **Session 1: Part 2**.

Session 1: Part 2

Hand back all student materials collected from the previous session. Have students review their investigation plan and make any adjustments, if necessary. Make extra copies of the model template page in the event students revise their model. Remind students to attach any revisions to the back of ***Student Answer Packet 1***.

8. Gather all the materials necessary to perform your planned investigation. Set up your investigation.
9. Complete and label the model in ***Student Answer Packet 1*** to show your setup with investigation variables. For each design change, a new model setup must be completed. Attach revised models to the back of ***Student Answer Packet 1***.
10. Working with your partner, conduct your planned investigation. Make sure to record all your numerical data for each trial in the data table for your **Investigation Option**. Also, record any relevant observations while conducting your investigation.
11. Clean up your investigation materials and return all materials to their original location.

Have designated areas for wet sediment to be returned.

12. Evaluate your collected data with your partner to make a claim using numerical data in ***Student Answer Packet 1*** about whether the data supports or refutes your prediction. (You will only need to make a claim about the **Investigation Option** you and your partner performed.)
13. Share your data table with the rest of the class, as directed by your teacher. Record the data from the other **Investigation Option** displayed by your teacher into the data table in ***Student Answer Packet 1***.

Teachers can determine how best to share student data (written or digital) with the goal that students have access to all data sets so students can choose one data set that has been communicated to the rest of the class for analysis. After reviewing all student data, teachers may want to direct students to data sets that show clear relationships between variables.

14. Indicate the students' names that you obtained your data from in the data table.

15. Gather all the materials necessary to perform **Investigation C**.
16. Predict the type of weathering that will most likely occur in this investigation. Identify one piece of observational evidence that would support this prediction.
17. Make a prediction about which type of liquid will cause the greatest amount of weathering to the chalk. Give a reason for your answer.
18. Use appropriate safety equipment and follow all safety procedures as determined by your teacher when working with the vinegar and chalk.

Students with chalk allergies should **not** handle chalk and may obtain this data from their partner. Remind students that goggles must be worn when working with vinegar, and tweezers should be used to handle chalk.

19. Working with your partner, you will collect data as evidence for how different liquids (tap water and vinegar) affect the weathering of an Earth material (calcium carbonate chalk).
20. Label one cup “vinegar” and one cup “tap water” using masking tape. The vinegar represents acid rain and the tap water represents normal precipitation.
21. Mass each piece of chalk and record this value in the data table, *Effect of Type of Liquid on Amount of Weathering*, in **Student Answer Packet 1** in the “Initial” column.
22. Place one piece of chalk in the bottom of each cup.
23. Cover each piece of chalk with equal amounts of the labeled liquids.
24. Begin timing and record your observations in the data table about changes to the chalk or liquid.
25. After 5 minutes, carefully remove each piece of chalk from the labeled liquid.
26. Mass each piece of chalk and record this value in the data table on **Student Answer Packet 1** in the “Final After 5 minutes” column.
27. Calculate the change in mass for each piece of chalk and record this value in the data table in **Student Answer Packet 1** in the “Change” column.
28. Evaluate your collected data with your partner to make a claim at the bottom of your data table in **Student Answer Packet 1** about whether the data supports or refutes your prediction.
29. Return your **Student Directions** and **Student Answer Packet 1** to your teacher.

Session 2

Return **Student Answer Packet 1** and **Student Directions** to the students, as they can be used to answer some **Student Answer Packet 2** questions.

The class data table for **Investigation C** should be displayed/available to students, along with all data sets from **Session 1: Part 2**, as they can also be used to answer some **Student Answer Packet 2** questions.

Your task in **Session 2** is to apply what you learned in **Session 1** to independently answer the investigation questions found in **Student Answer Packet 2**.

Answer/Scoring Key**Student Answer Packet 1**

Investigation Option A: An acceptable response must include a correctly related independent and dependent variable. Acceptable responses include, but are not limited to:

4a. **Independent Variable:** slope (height of stream table above surface)

Dependent Variable: distance *or* time *or* amount of sediment/figurines moved

Independent Variable: volume (amount of water falling on landscape)

Dependent Variable: distance *or* time *or* amount of sediment/figurines moved

Independent Variable: type of sediment

Dependent Variable: distance *or* time *or* amount of sediment/figurines moved

Independent Variable: permeability of surface (bare sediment *or* sediment covered with acetate *or* washcloth)

Dependent Variable: distance *or* time *or* amount of sediment/figurines moved

Investigation Option B: An acceptable response must include a correctly related independent and dependent variable. Acceptable responses include, but are not limited to:

4a. **Independent Variable:** chosen sediment(s) *or* material(s) (affecting permeability)

Dependent Variable: infiltration rate *or* amount of runoff (water remaining on top) *or* water retention (amount remaining inside rain garden design)

- 4b. **Prediction:** An acceptable response for an **Investigation Option**-specific prediction must include the effect of the independent variable on the dependent variable.

Acceptable predictions include, but are not limited to:

- As the slope of the stream table increases, the distance sediment travels will increase.
- A rain garden with different-sized sediments (sorted into layers) will retain the most water, which will reduce surface runoff.

6. **Investigation Plan** (step-by-step procedure): Acceptable procedures must include:

- title
- purpose
- description of data to be collected
- description of evidence to be derived from data
- appropriate materials
- step-by-step procedure (focused on purpose with specific steps allowing the **Investigation Option** to be replicated)

7. Acceptable table creation and data collection must include:

- quantifiable data for each Investigation Option
- three trials with averaged data
- appropriate labels
- title
- independent and dependent variables recorded for each Investigation Option

9. A complete and labeled acceptable model must include:

- all materials used
- a labeled independent variable

12. An acceptable student claim must include numerical data that supports or refutes the student's prediction.
16. An acceptable response identifies chemical weathering and one piece of observational evidence. Acceptable responses include, but are not limited to:
- Bubbling will be observed.
 - The vinegar will become cloudy.
 - The chalk will lose volume/mass.
17. An acceptable student prediction must include reasoning. Acceptable responses include, but are not limited to:
- Vinegar is an acid liquid/lower pH so it should probably weather the chalk more than just water.
 - Vinegar is like acid rain and will weather the chalk faster because the chalk is made of the same minerals as calcite, the same mineral that makes up limestone.
- 21, 26, 27. An acceptable data table must be correctly completed.

Note: If the mass of the chalk decreased, it must be indicated with a negative value.

28. An acceptable response must include a claim that supports or refutes the student's prediction using data from **Investigation C**.

Answer/Scoring Key**Student Answer Packet 2**

1. Acceptable responses must include the correct relationship between the independent and dependent variable referenced in the investigation plan and data table in ***Student Answer Packet 1***.

Acceptable responses for the relationship include, but are not limited to:

- As the slope (height of the stream table) increases, the distance sediment/figurines are transported increases.
- As the amount of water poured increases, the amount of sediment/figurines transported increases.

Note: Student responses must include *both* the correct relationship and supporting data as evidence.

2. Acceptable responses for the relationship include, but are not limited to:

- The cut bank in the photo is evidence of the destructive process of erosion, while the point bar in the photo is evidence of the constructive process of deposition.
- Water is moving faster on the outside, contributing to the destructive process of erosion, which creates a deeper, wider stream channel, while water is moving slower on the inside, contributing to the constructive process of deposition, which creates a shallower stream channel with less carrying capacity, resulting in the addition of sediment.

Note: Student responses must include a description of *both* constructive and deconstructive processes.

3. Acceptable responses must include the correct relationship between the independent and dependent variable referenced in the investigation plan and data table in ***Student Answer Packet 1***.

Acceptable responses for the relationship include, but are not limited to:

- Well-sorted, loosely packed sand will have a high infiltration rate, resulting in less surface runoff.
- When a layer with smaller particles (topsoil) is above a layer of larger particles (gravel), water infiltration is controlled, decreasing runoff to storm sewers.
- A layer of topsoil with plants will absorb water, and permeable gravel underneath will allow for infiltration into the groundwater.

Note: Student responses must include *both* the correct relationship and supporting data as evidence.

4a. Acceptable responses include, but are not limited to:

- Larger-sized particles have a greater permeability, therefore the gravel had greater permeability and the most water infiltrated/drained into the cup.
- Rain gardens have vegetation and gravel layers that are very permeable and that absorb water so it doesn't become runoff.

4b. Acceptable responses include, but are not limited to:

- A measured volume of sediment in the rain garden container (e.g. 100 ml) will need to have enough water poured into the container until it reaches the top of the sediment column (e.g. 50 ml). Subtract the volume of water added to the soil from the total volume of sediment to determine percent porosity (e.g. 50%).

4c. Acceptable responses include, but are not limited to:

- Effective rain garden materials (sediments) must be porous to allow infiltration and retain water at a controlled rate.

5. Acceptable responses include, but are not limited to:

- The presence of rain gardens increases surface vegetation in urban settings, reducing surface runoff during precipitation events.
- The presence of rain gardens causes less precipitation to become runoff because plant roots and sediment layers absorb and retain (infiltrate) stormwater.

6. Acceptable responses include, but are not limited to:

- The longer a mineral/chalk is exposed to the chemical weathering (vinegar) process, the more the mass of the original sample decreases.
- Acidic liquids will chemically weather chalk/limestone more than water alone.

7. Acceptable responses include, but are not limited to:

- Expansion of freezing water is a unique property that greatly influences the physical breakdown of rocks in New York State, which contributes to the ongoing formation of fallen rock zones along roadways in mountain regions (Adirondacks, Catskills, Taconics).
- The physical property of abrasion is caused by moving water carrying particles that contribute to the weathering of rocks/landscapes (falls, gorges, potholes, and gullies found in the Allegheny Plateau and the Erie-Ontario Lowlands).
- The chemical property of hydration dissolves certain rock types, like limestone and rock salt, which creates sinkholes and caverns in places like the Hudson Valley and the Allegheny Plateau.

Note: Student responses must include a variable that was *not* investigated and described in **Investigation C**.

8. Acceptable responses include, but are not limited to:

Genesee River:

Spatial:

- More land area is eroded and deposited along the Genesee River basin as a result of running water along the length of the river and its tributaries.
- The moving water of the Genesee River weathers, erodes, and deposits land along its entire 250 km length across New York State.

Temporal:

- Rapidly changing volumes of water cause rapid changes in land surfaces along the Genesee River.
- Water on Earth's surface changes the landscape at a faster rate than water flowing underground.

Allegheny Plateau Caves:

Spatial:

- The caves formed in limestone or other carbonate rock formations occur in localized areas.
- Cave formation is limited to small geographic areas containing carbonate bedrock.

Temporal:

- Acidic groundwater causes carbonate rock to slowly dissolve, forming underground caves.
- Infiltration of acidic groundwater and the process of dissolution of carbonate rock forms caves over a long period of time.

9. Mantle Rock *D*: Liquid

Mantle Rock *E*: Solid

Acceptable explanations include, but are not limited to:

- Mantle rock at location *D* contains water, causing it to melt at lower temperatures under the same pressure and depth conditions as rock without water.